ArbitrageIQ — Complete Technical Report

Real-Time Crypto Arbitrage Trading Terminal

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Category: Data Engineering / Trading Systems

Status: Live Operational / Operational

Live App: yemifatodu-arbitrageiq.streamlit.app

Repository: github.com/yemifatodu/arbitrageiq

1. Executive Summary

ArbitrageIQ is a production-grade cryptocurrency arbitrage scanning terminal that monitors 5 major exchanges (Binance, KuCoin, Bybit, OKX, Gate.io) across 30 cryptocurrency pairs in real-time. It calculates **true net profit after all fees** (trading + withdrawal) and delivers instant alerts via Telegram, with full historical tracking in a SQLite database.

The system uses an async architecture to fetch prices from all exchanges concurrently, completing 150 requests in ~150ms. It applies a minimum volume filter ($100k USD) to prevent slippage and implements geographic filtering to ensure accessibility from Nigeria.

> **Honest Caveat:** While the system detects arbitrage opportunities in real-time, **manual execution is too slow** to capture them profitably. Price discrepancies last seconds, not minutes. By the time a trader sees the Telegram alert and executes the trade, the opportunity is gone. The system is a decision-support tool and research platform, not an automated trading bot.

This report documents the system architecture, methodology, exploratory findings, performance metrics, and business implications, distinguishing clearly between what has been built versus what remains as future work.

2. Introduction

Cryptocurrency arbitrage exploits price discrepancies between exchanges — buying an asset where it's cheap and selling where it's expensive. In theory, this is risk-free profit. In practice, most retail traders lose money because they ignore trading fees, withdrawal costs, and execution latency.

This project builds a complete arbitrage scanning system that accounts for these real-world constraints and packages the detection logic into a usable decision-support tool.

3. Problem Statement

Crypto arbitrage opportunities exist when the same asset trades at different prices on different exchanges. But most "arbitrage bots" calculate **gross spread** (raw price difference) without accounting for:

- **Trading fees** (0.1% per trade × 2 = 0.2% total)
- **Withdrawal fees** (0.05% average, varies by exchange)
- **Slippage** (price movement during execution)
- **Latency** (time from detection to execution)

A "profitable" 0.25% spread becomes **breakeven or loss** after fees. Additionally, opportunities typically last less than 30 seconds, making manual monitoring impossible.

4. Business Background

Crypto arbitrage is a competitive, high-frequency activity. Professional quant firms use co-located servers, pre-funded accounts, and automated execution to capture opportunities in milliseconds. Retail traders cannot compete on speed, but they can compete on **information** — knowing where opportunities exist, even if they can't execute fast enough to capture them.

ArbitrageIQ serves as:

- **Educational tool** — understanding arbitrage mechanics and market microstructure
- **Research platform** — identifying which exchange pairs have persistent spreads
- **Alert system** — notifying traders of opportunities during high-volatility events
- **Foundation** — building block for automated execution systems

5. Objectives

Primary objective

Build a real-time arbitrage scanning system that detects price discrepancies across 5 exchanges and calculates **true net profit after all fees**.

Secondary objectives

- Deliver instant alerts via Telegram for immediate notification
- Persist historical opportunities for analytics and backtesting
- Build a professional trading terminal UI for real-time monitoring
- Implement geographic filtering for Nigeria-accessible exchanges
- Apply liquidity filters to prevent slippage on low-volume pairs

Success criteria / KPIs

KPI	Type	Target / Result
Scan time	Technical	<200ms for 150 concurrent requests (achieved: ~150ms)
Exchanges monitored	Technical	5 (achieved)
Symbols tracked	Technical	30 (achieved)
Min profit threshold	Technical	0.15% after fees (configurable)
System uptime	Technical	24/7 background scanner (achieved)

6. Target Users

- **Retail traders** — consumers of real-time alerts and market research
- **Quant developers** — consumers of the async architecture and fee calculation framework

- **Data engineers** — consumers of the real-time data pipeline design
- **Myself** — learning async Python, trading system architecture, and market microstructure

7. Dataset (Data Sources)

Source and size

Live market data from 5 exchanges via CCXT library: Binance, KuCoin, Bybit, OKX, Gate.io.

30 cryptocurrency pairs tracked per scan cycle: BTC, ETH, BNB, SOL, XRP, ADA, DOT, DOGE, AVAX, MATIC, LINK, NEAR, INJ, ARB, OP, APT, SUI, TRX, ATOM, FIL, LTC, SAND, MANA, AXS, AAVE, CRV, SNX, UNI, MKR, ETC.

Data quality

- **Missing values:** None (CCXT returns null for unavailable pairs, filtered out)
- **Duplicate records:** None (each scan is unique timestamp)
- **Volume filter:** Minimum $100k USD quote volume to ensure liquidity
- **Safe symbols whitelist:** 30 verified symbols to avoid ccxt.BadSymbol errors

Potential bias and ethical considerations

The system is biased toward **Nigeria-accessible exchanges** — Kraken and Coinbase are excluded due to IP blocking and geo-restrictions. This means the system may miss arbitrage opportunities available in other regions. Additionally, the system uses **REST polling** instead of WebSocket subscriptions, introducing latency that professional systems don't have.

8. Methodology

The project follows a real-time data engineering workflow:

- Background scanner runs every 5 minutes (configurable)
- Async price fetching from all 5 exchanges concurrently
- Arbitrage detection — find min/max prices across exchanges
- Fee calculation — subtract trading + withdrawal fees
- Threshold filtering — only save opportunities above min profit
- Database persistence — save to SQLite with indexed queries
- Telegram alerts — send formatted notifications
- Streamlit dashboard — display live opportunities and analytics

9. Data Cleaning (Data Filtering)

Because the system uses live market data, there's no traditional "data cleaning" step. However, several filtering steps ensure data quality:

- **Null price filtering** — pairs with no last price are skipped
- **Volume filtering** — pairs with <$100k quote volume are skipped (slippage protection)
- **Safe symbols whitelist** — only 30 verified symbols are scanned (avoid bad pair errors)
- **Exchange validation** — only Nigeria-accessible exchanges are used

10. Exploratory Data Analysis (Market Insights)

Overall opportunity rate

Across all scans, approximately **2-3% of scans find opportunities above 0.15% net profit threshold**.

Summary statistics (from historical data)

Metric	Mean	Std Dev	Min	Max
Net Profit %	0.18%	0.12%	0.15%	0.85%
Spread %	0.43%	0.28%	0.25%	1.10%
Fees %	0.25%	0.00%	0.25%	0.25%
Volume (USD)	$2.4M	$3.1M	$100k	$18M

Opportunity rate by exchange pair

Buy Exchange	Sell Exchange	Opportunity Count	Avg Net Profit
Binance	KuCoin	68% of all opps	0.22%
Binance	Gate.io	18% of all opps	0.19%
Binance	Bybit	8% of all opps	0.17%
KuCoin	Binance	4% of all opps	0.16%
Other	Other	2% of all opps	0.15%

Honest caveat: Binance is the "buy" exchange in 68% of opportunities because it consistently has the lowest prices (highest liquidity drives prices down). KuCoin and Gate.io are the "sell" exchanges because they have wider spreads. This pattern holds across 80% of all detected arbitrage routes.

Fee impact analysis

Gross Spread	Net Profit (after fees)	Viable?
0.25%	0.00%	No (breakeven)
0.30%	0.05%	Marginal
0.40%	0.15%	Yes (meets threshold)
0.50%	0.25%	Yes (good opportunity)
1.00%	0.75%	Yes (rare, high-profit)

12. System Architecture

```
[ Background Scanner ] -> [ Async Price Fetching via CCXT ] | [ SQLite Database ] <- [
Fee-Adjusted Net Profit Check ] | v [ Telegram Alerts & Streamlit UI ]
```

Async architecture

```
async def get_prices_batch(self, symbols: List[str]) -> Dict:
    tasks = [
        self.fetch_price(exchange, symbol)
        for exchange in self.exchanges
        for symbol in symbols
    ]
    responses = await asyncio.gather(*tasks, return_exceptions=True)
```

Fee configuration

```
TRADING_FEES = {
    'binance': 0.001, 'kucoin': 0.001, 'bybit': 0.001, 'okx': 0.001, 'gate': 0.001,
}
WITHDRAWAL_FEES = {
    'binance': 0.0005, 'kucoin': 0.0005, 'bybit': 0.0005, 'okx': 0.0005, 'gate': 0.0005,
}
```

Database schema

```
CREATE TABLE opportunities (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp TEXT, symbol TEXT, buy_exchange TEXT, sell_exchange TEXT,
    buy_price REAL, sell_price REAL, spread_pct REAL, fees_pct REAL,
    net_profit_pct REAL, profit_per_btc REAL, volume REAL
);
CREATE INDEX idx_timestamp ON opportunities(timestamp DESC);
CREATE INDEX idx_symbol ON opportunities(symbol);
```

13. Evaluation & Results

System performance

Metric	Value
Avg scan time	~150ms
Exchanges monitored	5
Symbols tracked	30
Concurrent requests	150
Min breakeven threshold	0.15% (after fees)
System uptime	24/7

14. Business Insights

- **Binance is consistently the cheapest exchange** (68% of opportunities) — highest liquidity drives prices down
- **KuCoin and Gate.io are consistently the most expensive** — wider spreads create sell opportunities
- **Mid-cap coins show higher profit but lower liquidity** — tradeoff between profit and execution risk
- **Asian trading hours (2am-8am UTC) have more opportunities** — liquidity fragmentation across exchanges

18. Conclusion

ArbitrageIQ demonstrates a complete, production-grade arbitrage scanning system — from async multi-exchange price fetching to fee-adjusted profit calculations and real-time Telegram alerts.

20. References

- CCXT library documentation (https://docs.ccxt.com/)
- Streamlit documentation (https://docs.streamlit.io/)
- Telegram Bot API documentation (https://core.telegram.org/bots/api)